Systems Analysis and Control

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Lecture 11: Proportional, Differential, and Integral Feedback

Overview

In this Lecture, you will learn:

Limits of Proportional Feedback $u(t) = ke(t)$

- Performance Specifications.

Derivative Feedback $u(t) = k_D \dot{e}(t)$

- Pros and Cons
- PD Control
- Pole Placement

More on Steady-State Error

- Response to ramps and parabolae
- Limits of PD control

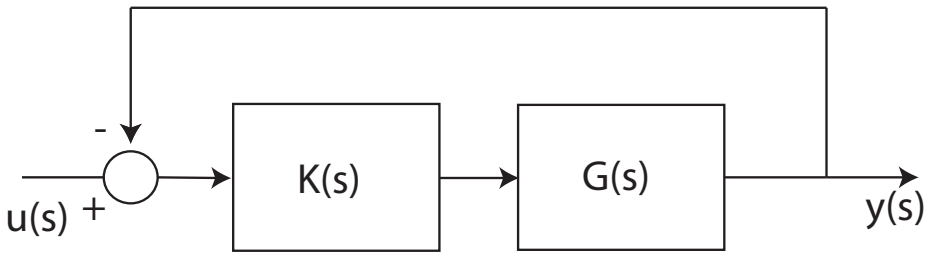
Integral Feedback $u(t) = \int_t k_I e(t) dt$

- Elimination of steady-state error
- Pole-Placement

Proportional (P) Feedback: Limitations

Proportional Feedback cannot meet any performance specs

- **Proportional Feedback:** $K(s) = k$
  $u(t) = ke(t) = k(u(t) - y(t))$
- **Plant:** $\hat{y}(s) = \hat{G}(s)\hat{u}(s)$



Recall Inverted Pendulum:

$$\hat{G}(s) = \frac{1}{s^2 - \frac{Mgl}{2}}$$

Closed Loop Transfer Function :

$$\frac{GK}{1+GK} = \frac{k}{s^2 - \frac{Mgl}{2} + k}$$

Closed Loop Pole Locations :

$$p = \begin{cases} \pm\sqrt{(\frac{Mgl}{2}) - k} & k < \frac{Mgl}{2} \\ \pm\imath\sqrt{k - \frac{Mgl}{2}} & k > \frac{Mgl}{2} \end{cases}$$

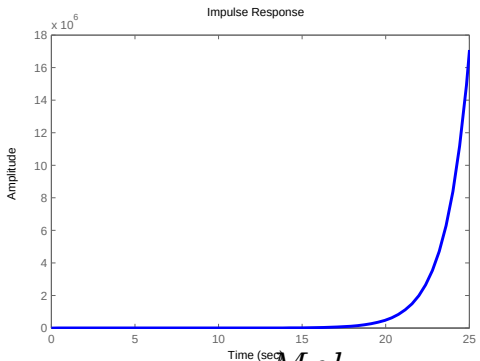


Figure: Case 2: $k < \frac{Mgl}{2}$ – UNSTABLE

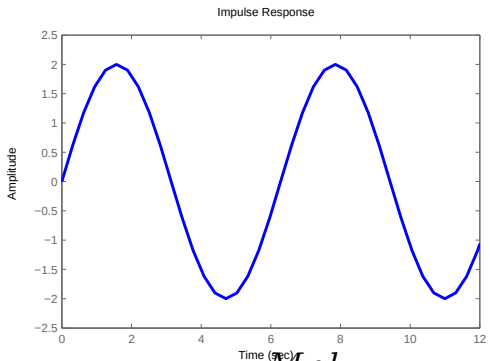
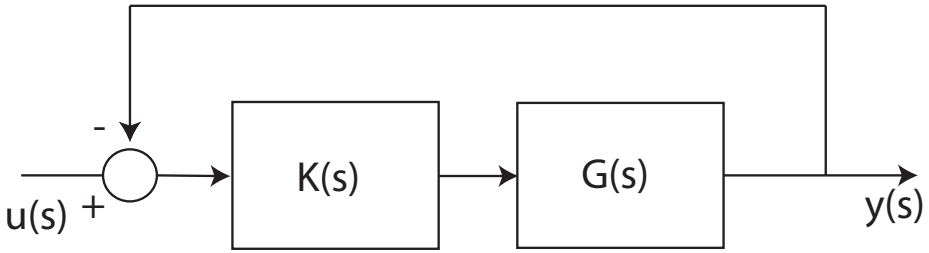


Figure: Case 1: $k > \frac{Mgl}{2}$ – UNDAMPED

Derivative (D) Feedback: Responding to Rate of Change

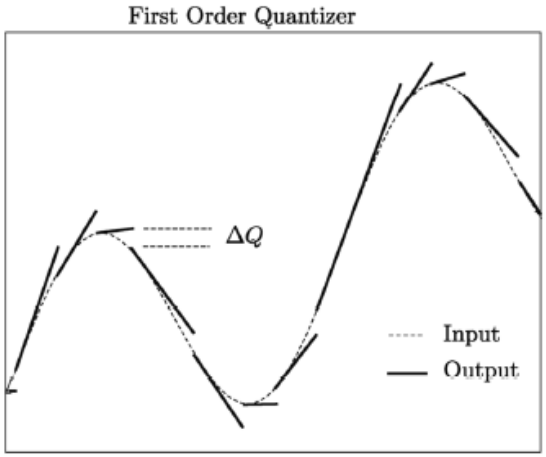
Derivative Feedback: $K(s) = k_D s$

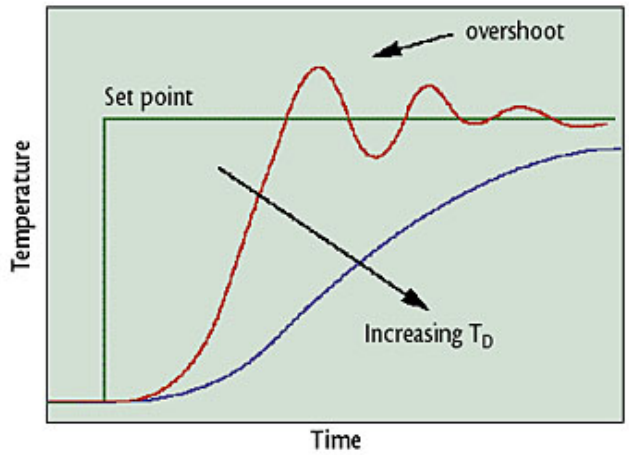
$$u(t) = k_D \dot{e}(t) = k_D(\dot{u}(t) - \dot{y}(t))$$



Derivative feedback measures the Rate of Change of the error.

- Proportional feedback responds to the Magnitude of the error.





Prediction: To measure $\dot{e}(t)$, recall the definition of derivative:

$$\dot{e}(t) \cong \frac{e(t + \Delta t) - e(t)}{\Delta t}$$

- The $\dot{e}(t)$ term depends on both the current position and predicted position.

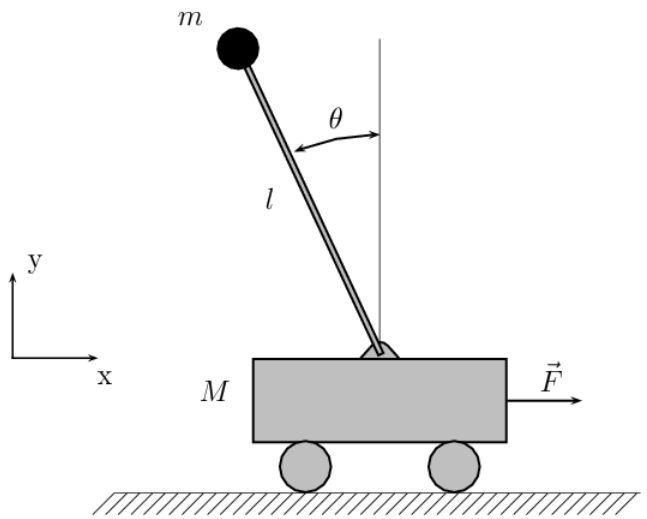
Problem: Derivative Feedback Ignores Magnitude of Error

Useless for Inverted Pendulum

Controller: $\hat{K}(s) = T_D s$

Closed Loop Transfer Function:

$$\frac{T_D/Js}{s^2 + (T_D/J)s - \frac{Mgl}{2J}}$$



2nd-Order System This is stable *if and only* if *both*

- $T_D/J > 0$
  and
- $-\frac{Mgl}{2J} > 0$

Derivative Feedback **Alone** cannot stabilize a system.

Proportional-Derivative (PD) Control

Differential Control is usually combined with proportional control.

- To improve stability (proportional part)
- To improve overshoot spec (derivative part)
- To reduce steady-state error (proportional part)

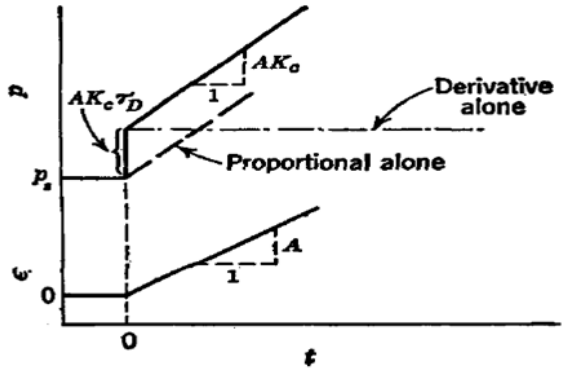


Figure: Proportional and Derivative Response to Ramp input

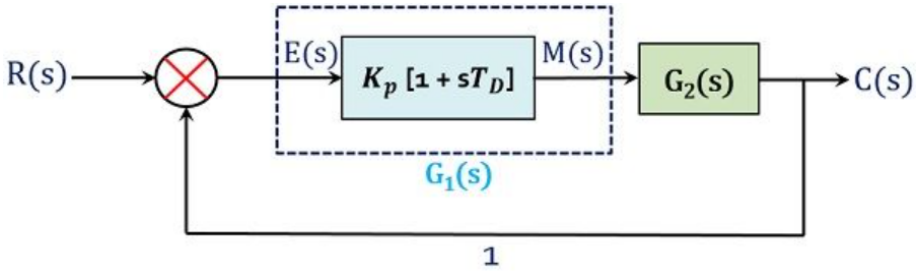


Figure: Closed-Loop PD controller

Proportional-Derivative (PD) Controller: The form of control is

$$u(t) = K\left[e(t) + T_D\dot{e}(t)\right]$$

or in the Laplace domain:

$$\hat{u}(s) = K\left[1 + T_D s\right]\hat{e}(s)$$

PD Control - Pole Placement

Applied to a 2nd-order system

Lets look at the effect of PD control on a 2nd-order system:

$$\hat{G}(s) = \frac{1}{s^2 + bs + c}$$

Controller: $\hat{K}(s) = K\left[1 + T_D s\right]$
Closed Loop Transfer Function:

$$\begin{aligned}\frac{\hat{K}(s)\hat{G}(s)}{1 + \hat{K}(s)\hat{G}(s)} &= \frac{K\left[1 + T_D s\right]}{s^2 + bs + c + K\left[1 + T_D s\right]} \\ &= \frac{K\left[1 + T_D s\right]}{s^2 + (b + KT_D)s + (c + K)}\end{aligned}$$

The poles of the system are freely assignable for a 2nd order system.

- The *Gains* T_D and K allow us to construct any denominator we desire.

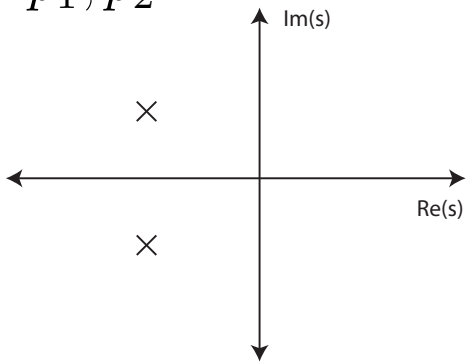
Generic PD Control - Pole Placement

Closed-Loop PD control applied to a 2nd-order system

Recall: Closed-loop transfer function:

$$\frac{\hat{K}(s)\hat{G}(s)}{1+\hat{K}(s)\hat{G}(s)} = \frac{K\left[1+T_D s\right]}{s^2+(b+KT_D)s+(c+K)}$$

Suppose we want poles at $s = p_1, p_2$.



- We want the closed loop of the form:

$$\frac{n(s)}{(s-p_1)(s-p_2)} = \frac{n(s)}{s^2-(p_1+p_2)s+p_1p_2}$$

Thus we want

- $c+K = p_1p_2$ which means $K = p_1p_2 - c$.
- $b+KT_D = -(p_1+p_2)$ which means $T_D = -\frac{p_1+p_2+b}{K} = -\frac{p_1+p_2+b}{p_1p_2-c}$

PD feedback gives Total Control over the poles of a 2nd-order system.

Generic PD Control Example

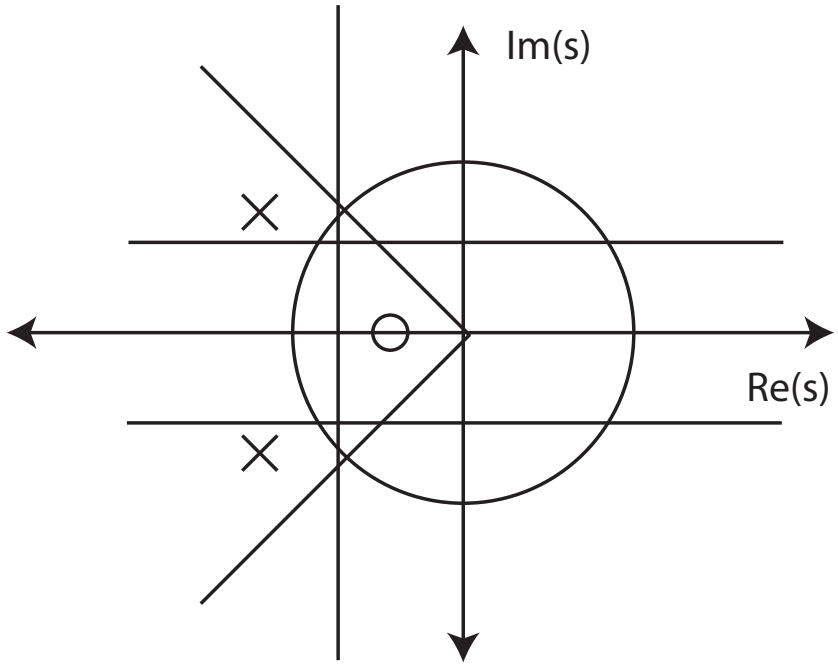
Pole Placement: Meet Performance Specs

Suppose we have the 2nd-order system

$$\hat{G}(s) = \frac{1}{s^2 + s + 1}$$

and performance specifications:

- **Overshoot:** $M_{p,desired} = .05$
- **Rise Time:** $T_{r,desired} = 1s$
- **Settling Time:** $T_{s,desired} = 3.5s.$



As we found in Lecture 9, these specifications mean that the poles satisfy:

$$\sigma < -.9535\omega, \qquad \sigma < -1.333, \qquad \omega_n > 1.8$$

We chose the pole locations:

$$p = -1.5 \pm 1.4\imath$$

Generic PD Control Example

Pole Placement: Determine gains K and T_D using selected poles

To get the pole locations: $p_{1,2} = -1.5 \pm 1.4\imath$, we want to equate the denominators (here $b = c = 1$)

$$\frac{n(s)}{s^2 - (p_1 + p_2)s + p_1 p_2} = \frac{K\left[1 + T_D s\right]}{s^2 + (b + KT_D)s + (c + K)}$$

we choose

- The Proportional Gain (K):

$$K = p_1 p_2 - c = (-1.5 + 1.4\imath)(-1.5 - 1.4\imath) + 1 = 1.5^2 + 1.4^2 - 1 = 3.21$$

- The Derivative Gain (T_D)

$$T_D = -\frac{p_1 + p_2 + b}{K} = -\frac{-3 + 1}{3.21} = \frac{2}{3.21} = .623$$

This gives the controller:

$$\hat{K}(s) = K(1 + T_D s) = 3.21 + 2s$$

PD Control has NO effect on Steady-State Error

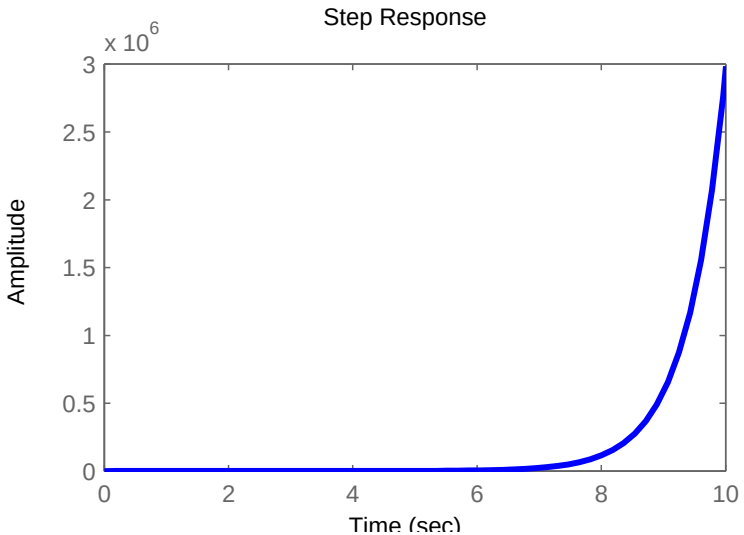


Figure: Open Loop

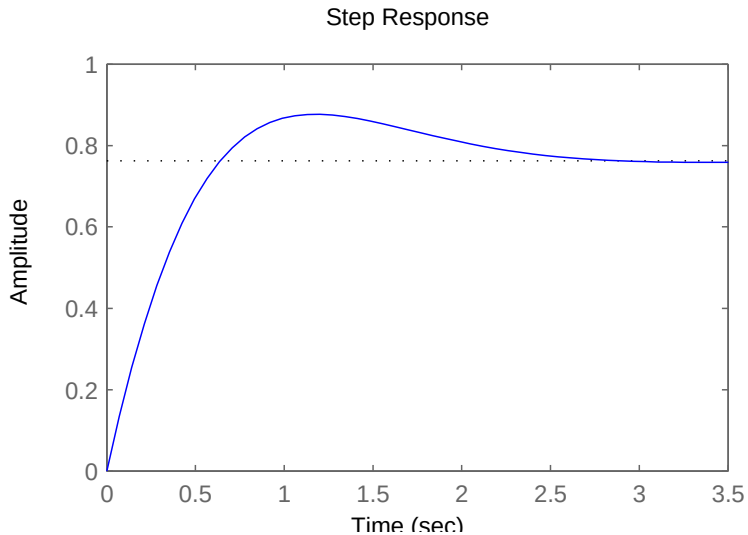


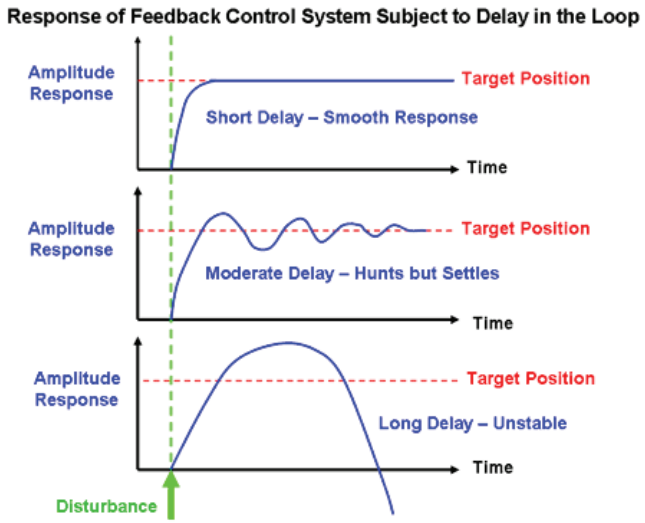
Figure: Closed Loop

Although the PD controller gives us control of the pole locations, the steady-state value is

$$y_{ss} = \frac{K}{c+K} = \frac{3.21}{4.21} = .7625$$

Another Problem with Derivative Control

Implementation using Delay (Dangerous!)



Problem: Differential control is implemented using delay.

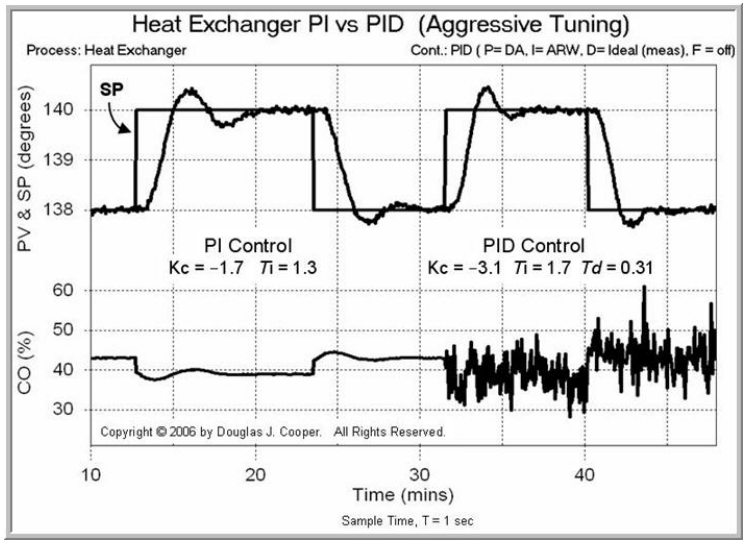
- $y(t)$ is the measurement.
- $\dot{y}(t)$ cannot be measured directly
  - Approximate using the delayed response:

$$\dot{y}(t) \cong \frac{y(t) - y(t - \Delta t)}{\Delta t}$$

  - Delay can cause instabilities.

Yet Another Problem with Derivative Control

Amplifies Sensor Noise (Dangerous!)



Noise Amplification: Feedback Control requires a sensor (which has noise)

- Measurement of $\dot{y}(t)$ is heavily influenced by noise.

$$\dot{y}(t) \cong \frac{y(t) - y(t - \Delta t)}{\Delta t}$$

- Sensor measurements have error $(\tilde{y} = y \pm \sigma)$
- As $\Delta t \to 0$, the effect of noise, σ is amplified:

$$\dot{\tilde{y}}(t) = \frac{y(t) - y(t - \Delta t)}{\Delta t} + \frac{2\sigma}{\Delta t} \to \infty$$

Integral Control is Used to Eliminate Steady-State Error

Problem: Everyone hates Steady State Error

- Proportional (and PD) Control ALWAYS has steady-state error.
- Small persistent errors result in small actuator response.

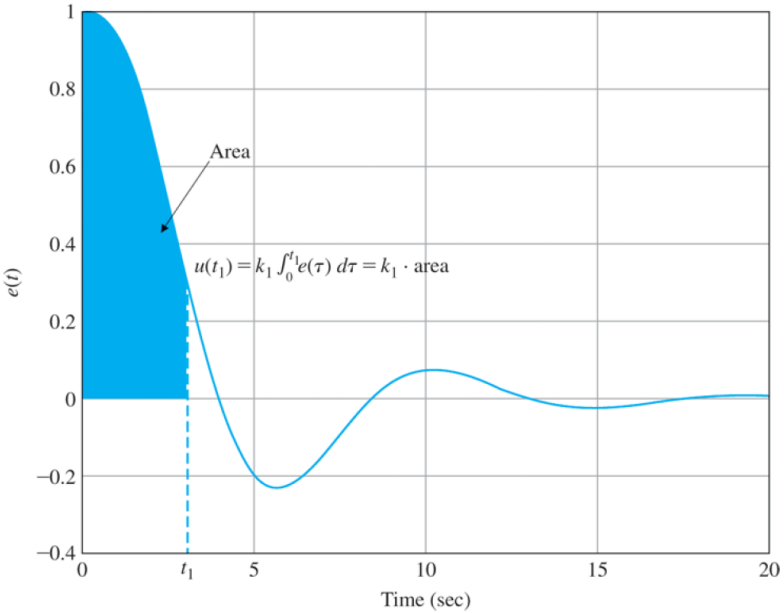
Integral control amplifies the effect of small persistent errors by integrating them over time.

Integral Controller: Control effort is

$$u(t) = \frac{1}{T_I}\int_0^t e(\theta)d\theta$$

or, in the Laplace transform

$$\hat{u}(s) = \frac{1}{T_I s}\hat{e}(s)$$



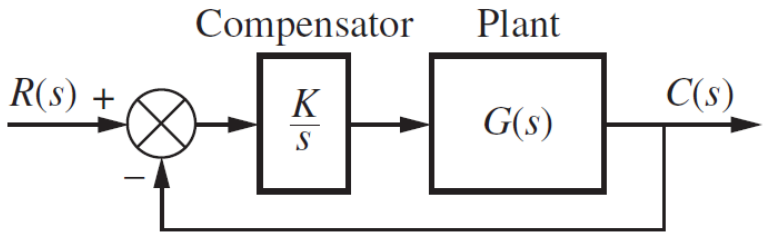
One must be careful when using integral feedback

- Adds a pole at the origin.
- Creates **Windup** (blows up) if model is wrong or error is uncontrollable/incorrectly measured

Integral Control is Often Destabilizing

Closing the Loop: $\hat{K}(s) = \frac{1}{T_I s}$

$$\frac{\hat{G}(s)\hat{K}(s)}{1+\hat{G}(s)\hat{K}(s)} = \frac{\hat{G}(s)}{T_I s + \hat{G}(s)}$$



or if $G(s) = \frac{n_G(s)}{d_G(s)}$

$$\frac{\hat{G}(s)\hat{K}(s)}{1+\hat{G}(s)\hat{K}(s)} = \frac{n_G(s)}{T_I s \cdot d_G(s) + n_G(s)}$$

Suspension System Example: $G(s) = \frac{s^2+s+1}{s^4+2s^3+3s^2+s+1}$

$$\frac{\hat{G}(s)\hat{K}(s)}{1+\hat{G}(s)\hat{K}(s)} = \frac{s^2+s+1}{T_I s^5 + 2T_I s^4 + 3T_I s^3 + (T_I+1)s^2 + (T_I+1)s + 1}$$

If we set $T_I = .1$, then the transfer function has poles at

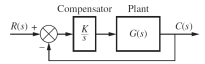
- $p_{1,2} = -.55 \pm .89\imath$, $p_3 = -2.26$, $p_{4,5} = .6384 \pm 1.877\imath$

Integral feedback can Destabilize the system where proportional feedback couldn't!

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Integral feedback can Destabilize the system where proportional feedback couldn't!

Integral Control ALWAYS eliminates steady state error to step input. This is because

$$\frac{\hat{G}(s)\hat{K}(s)}{1+\hat{G}(s)\hat{K}(s)}\Big|_{s=0} = \frac{n_G(s)}{T_I s \cdot d_G(s) + n_G(s)}\Big|_{s=0} = \frac{n_G(0)}{n_G(0)} = 1$$

Combine Integral Control with Proportional Control (PI)

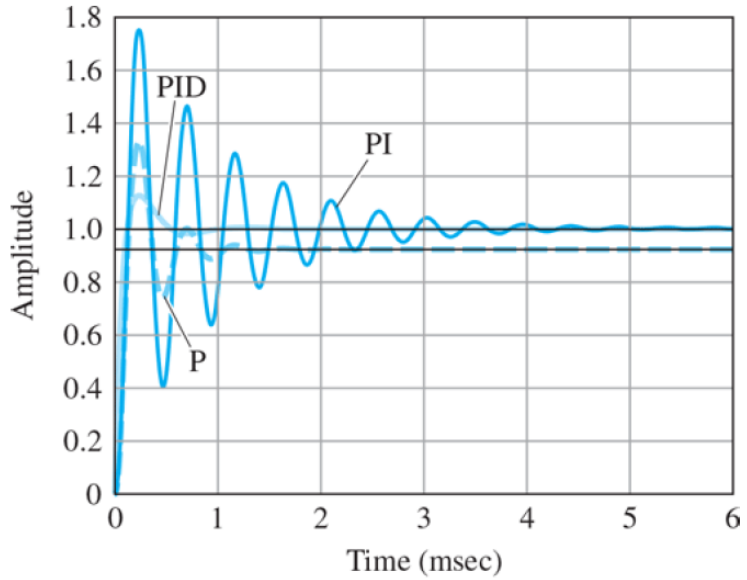
And Sometimes with Derivative Control (PID)

Integral Feedback Alone is destabilizing!

PI Feedback: Proportional-Integral

$$u(t) = K\left(e(t) + \frac{1}{T_I}\int_0^t e(\theta)d\theta\right)$$

$$\hat{K}(s) = K\left(1 + \frac{1}{T_I s}\right)$$

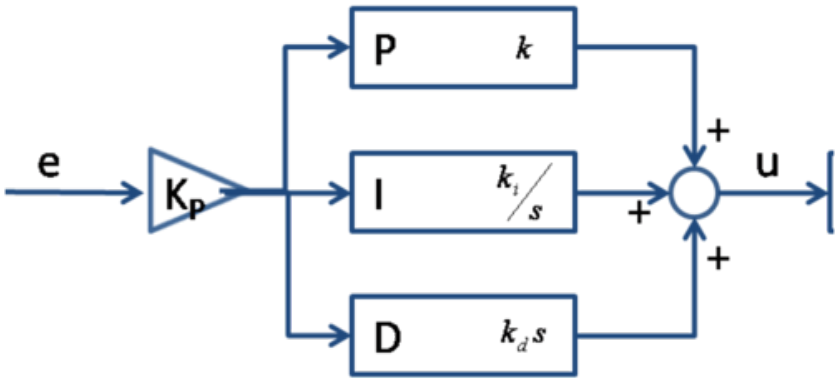


PID Feedback:

Proportional-Integral-Differential

$$u(t) = K\left(e(t) + \frac{1}{T_I}\int_0^t e(\theta)d\theta + T_D\dot{e}(t)\right)$$

$$\hat{K}(s) = K\left(1 + \frac{1}{T_I s} + T_D s\right)$$



PID ALWAYS eliminates steady state error to step input

Define Plant and PID Controller:

$$\hat{G}(s) = \frac{n_G(s)}{d_G(s)} \qquad \hat{K}(s) = K\left(1 + \frac{1}{T_I s} + T_D s\right) = \frac{K(T_D s^2 + s + T_I)}{s} = \frac{n_K(s)}{d_K(s)}$$

PID control always eliminates steady state error to a step input because

$$y_{ss} := \left.\frac{\hat{G}(s)\hat{K}(s)}{1 + \hat{G}(s)\hat{K}(s)}\right|_{s=0} = \left.\frac{n_G(s)n_K(s)}{s \cdot d_G(s) + n_G(s)n_K(s)}\right|_{s=0} = \frac{n_G(0)n_K(0)}{n_G(0)n_K(0)} = 1$$

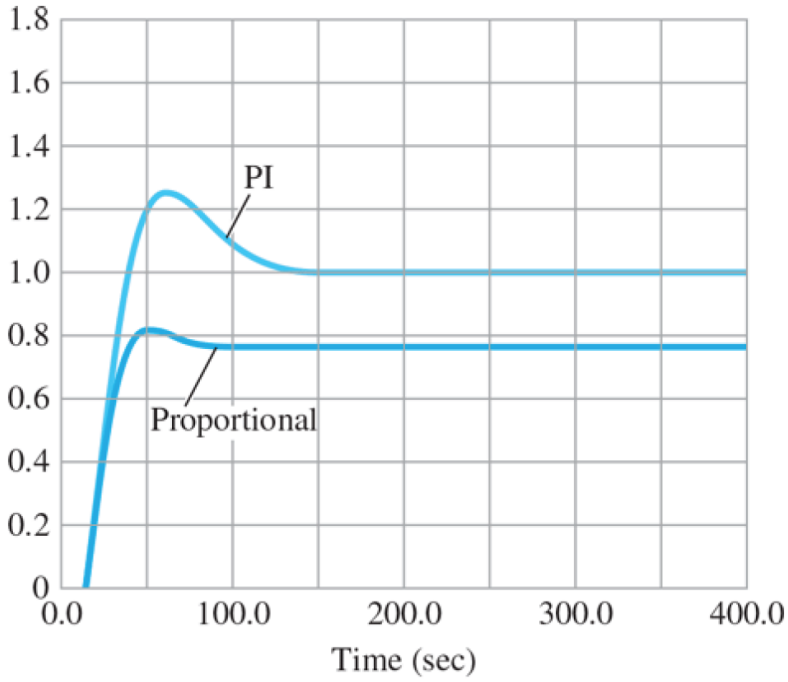
However: Integral Control will:

- Increase Settling Time (T_s)
- Increase Percent Overshoot (M_p)

These effects can be mitigated using by increasing derivative gain, T_D

- In practice, tuning the gain parameters is more art than science

The Major Benefit of PID: You don't really need a good model to tune PID controllers



Pole Placement using PID Control

Finally, lets see the effect of PID control on a second-order system:

$$\hat{G}(s) = \frac{1}{s^2 + bs + c}$$

Closed Loop:

$$\frac{\hat{G}\hat{K}}{1 + \hat{G}\hat{K}} = \frac{KT_D s^2 + Ks + K\frac{1}{T_I}}{s^3 + (b + KT_D)s^2 + (c + K)s + \frac{K}{T_I}}$$

Pole Placement: The three pole locations can be determined exactly.

- Given three poles: p_1, p_2, p_3.
- Construct Desired denominator:

$$\frac{1}{(s - p_1)(s - p_2)(s - p_3)} = \frac{1}{s^3 + a_d s^2 + b_d s + c_d}$$

Three equations:

- $b + KT_D = a_d$
- $c + K = b_d$
- $\frac{K}{T_I} = c_d$

Which can be solved as

- $K = b_d - c$
- $T_I = \frac{K}{c_d}$
- $T_D = \frac{a_d - b}{K}$

Summary

What have we learned today? In this Lecture, you learned:

Limits of Proportional Feedback

- Performance Specifications.

Derivative Feedback

- Pros and Cons
- PD Control
- Pole Placement

More on Steady-State Error

- Response to ramps and parabolae
- Limits of PD control

Integral Feedback

- Elimination of steady-state error
- Pole-Placement

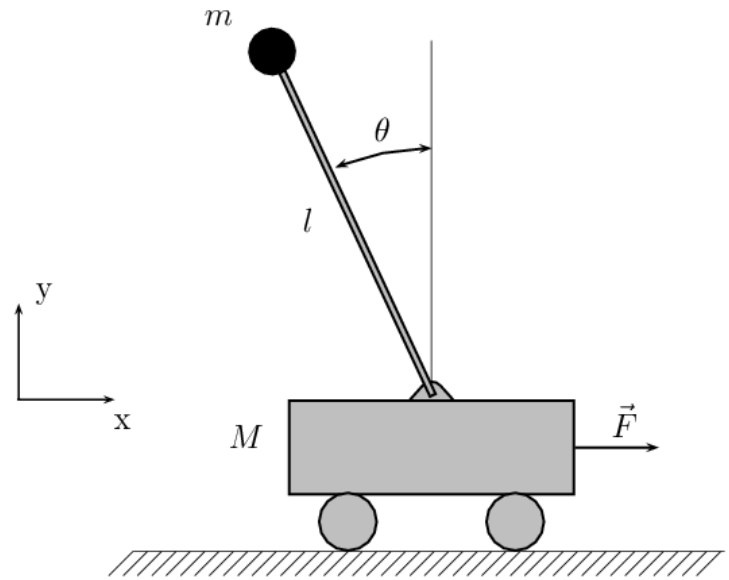
Yet Another PD Control Example

Inverted Pendulum Revisited

Lets look at the effect of PD control on the inverted Pendulum:

$$\hat{G}(s) = \frac{1/J}{s^2 - \frac{Mgl}{2J}}$$

Controller: $K\left[1 + T_D s\right]$



Closed Loop Transfer Function:

$$\frac{\hat{K}(s)\hat{G}(s)}{1 + \hat{K}(s)\hat{G}(s)} = \frac{K/J\left[1 + T_D s\right]}{s^2 - \frac{Mgl}{2J} + K/J\left[1 + T_D s\right]}$$

$$= \frac{K/J\left[1 + T_D s\right]}{s^2 + K/JT_D s + (K/J - \frac{Mgl}{2J})}$$

Yet Another PD Control Example

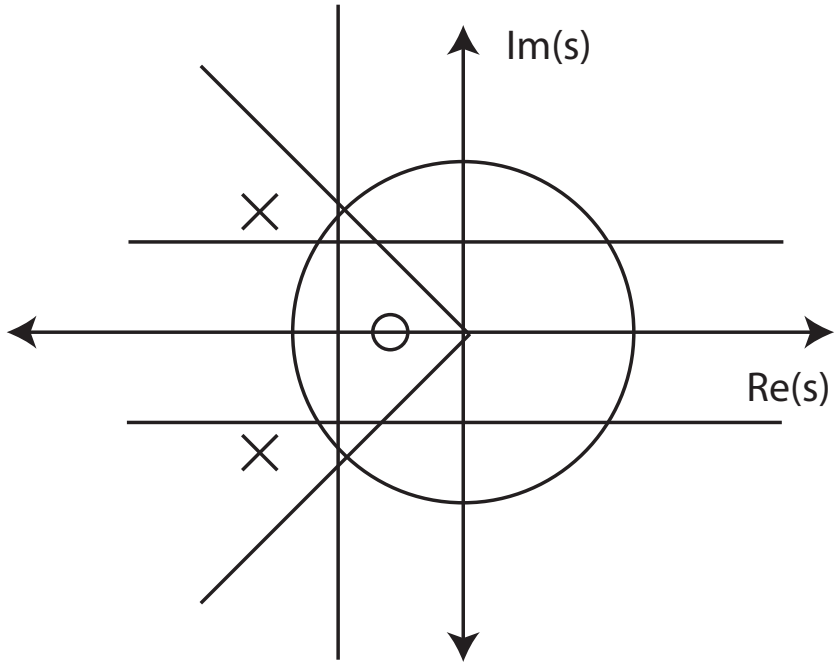
Inverted Pendulum : Desired Pole Locations

To achieve the performance specifications:

- **Overshoot:** $M_{p,desired} = .05$
- **Rise Time:** $T_{r,desired} = 1s$
- **Settling Time:** $T_{s,desired} = 3.5s.$

We want poles at

$$s = -1.5 \pm 1.4\imath$$



Thus we want

- $c + K = p_1 p_2$ which means $K = p_1 p_2 - c.$
- $b + KT_D = -(p_1 + p_2)$ which means

$$T_D = -\frac{p_1 + p_2 + b}{K} = -\frac{p_1 + p_2 + b}{p_1 p_2 - c}$$

Yet Another PD Control Example

Inverted Pendulum: Selecting Gains

The closed loop is

$$\frac{K/J\left[1+T_D s\right]}{s^2+K/JT_D s+(K/J-\frac{Mgl}{2J})}$$

To get the pole locations $p_{1,2}=-1.5\pm 1.4\imath$
we choose

- The Proportional Gain (K):

$$K/J=p_1p_2-c=4.21+\frac{Mgl}{2J}$$

- The Derivative Gain (T_D):

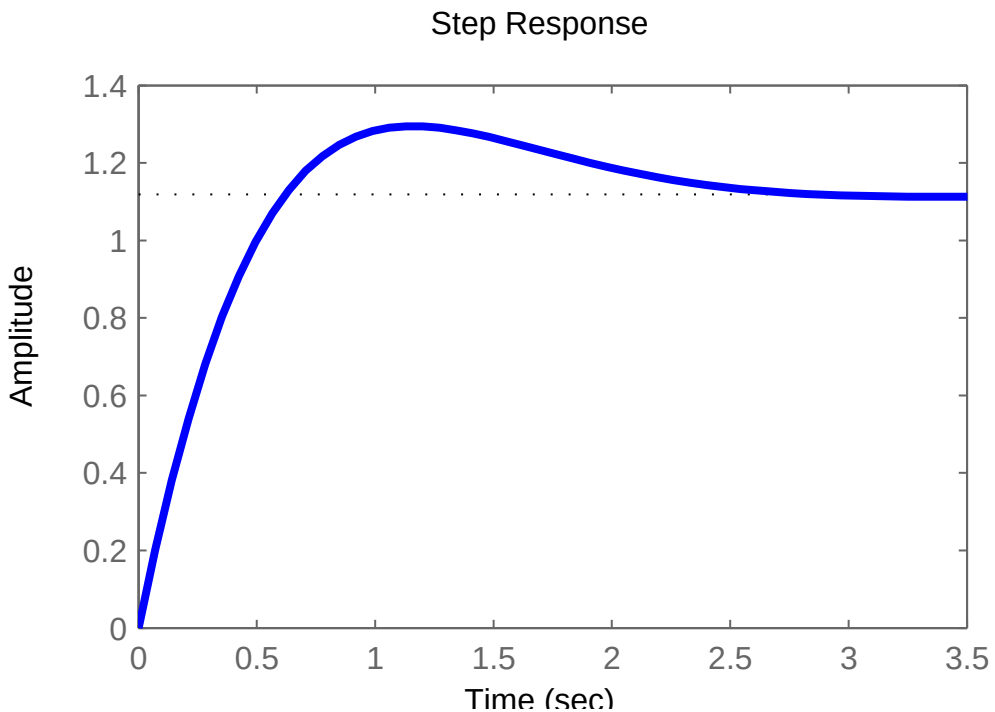
$$T_D=-\frac{p_1+p_2+b}{p_1p_2-c}=\frac{3}{4.21+\frac{Mgl}{2J}}$$

This gives the controller:

$$\hat{K}(s)=K(1+T_D s)=4.21J+\frac{1}{2}Mgl\left(1+\frac{3}{4.21+\frac{Mgl}{2J}}s\right)$$

Yet Another PD Control Example

Inverted Pendulum: Good Performance, but No Effect on Steady-State Error



The steady-state error with this controller is ($K = J = M = g = l = 1$)

$$y_{ss} = \frac{K/J}{(K/J - \frac{Mgl}{2J})} = \frac{4.21}{4.21 - .5} = 1.135$$

Derivative Control has **No Effect** on the steady-state error!

Recall: Steady-State Error

Lets take another look at steady-state error

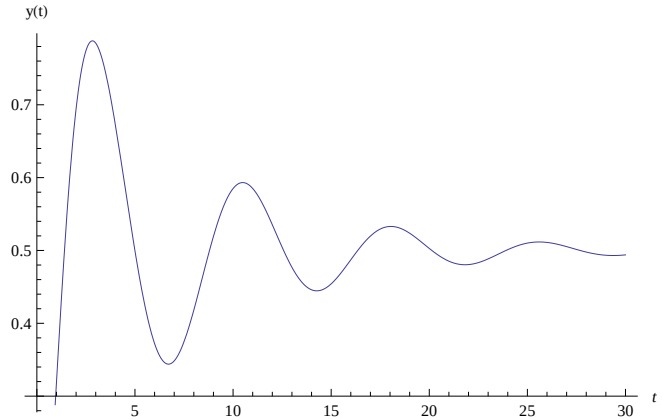


Figure: Suspension Response for $k = 1$

Problems:

- If target is moving, we may never catch up.
- Even if we can catch a moving target, we may not catch an accelerating target.

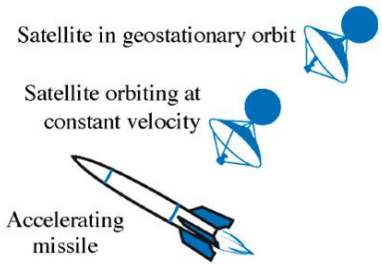
For these problems, the step response is not appropriate.

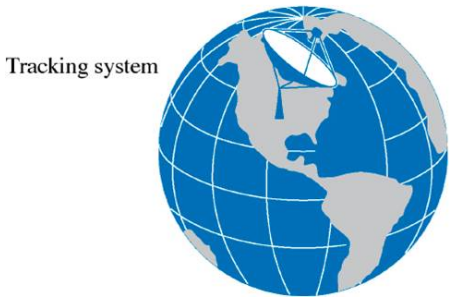
Recall:

- We measured steady-state error using the step response.
  - $e_{ss} = 1 - \lim_{t\to\infty} y(t)$

Sometimes this doesn't work.

- Assumes objective doesn't move.

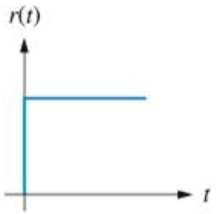
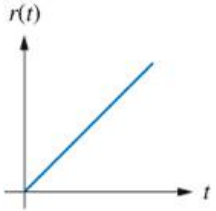
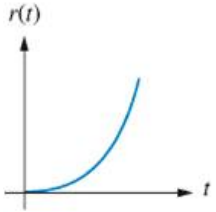




Ramp and Parabolic Inputs

There are other types of response we can consider.

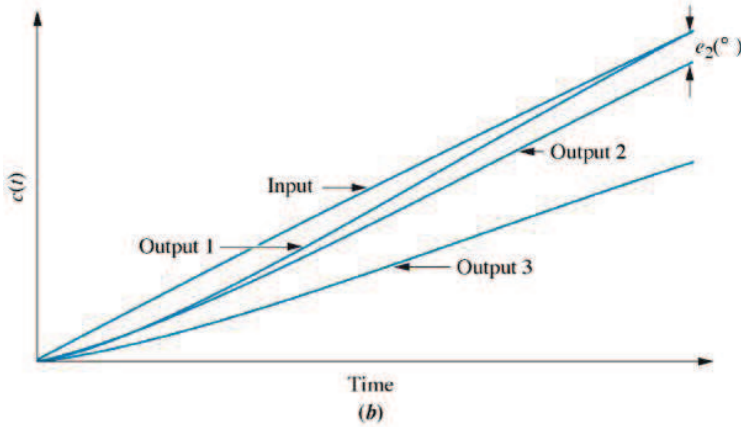
TABLE 7.1 Test waveforms for evaluating steady-state errors of position control systems

Waveform	Name	Physical interpretation	Time function	Laplace transform
	Step	Constant position	1	$\frac{1}{s}$
	Ramp	Constant velocity	t	$\frac{1}{s^2}$
	Parabola	Constant acceleration	$\frac{1}{2}t^2$	$\frac{1}{s^3}$

- Ramp response tracks error for a target with constant velocity.
- Parabolic response tracks error for a target with a constant acceleration.

Ramp and Parabolic Inputs

We can use the final value theorem to find the response to ramp and parabolic inputs:



Ramp Response:
Recall the ramp input:

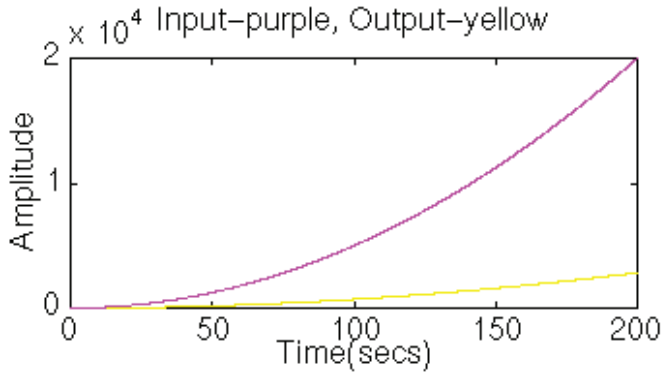
$$u(t) = t \qquad \hat{u}(s) = \frac{1}{s^2}$$

The steady-state error of $\hat{G}$ to a ramp input is

$$e_{ss} = \lim_{s\to 0} s\hat{e}(s) = \lim_{s\to 0} s(1-\hat{G}(s))\hat{u}(s) = \lim_{s\to 0} \frac{1-\hat{G}(s)}{s}$$

Ramp and Parabolic Inputs

We can use the final value theorem to find the response to parabolic inputs:



Parabolic Response:
Recall the parabolic input:

$$u(t) = t^2 \qquad \hat{u}(s) = \frac{1}{s^3}$$

The steady-state error in response of $\hat{G}$ to a parabolic input is

$$e_{ss} = \lim_{s\to 0} s(\hat{u}(s) - \hat{y}(s)) = s(1 - \hat{G}(s))\hat{u}(s) = \frac{1 - \hat{G}(s)}{s^2}$$

Note: The steady-state error to a parabolic input is usually infinite.

Ramp and Parabolic Inputs

The effect of the numerator

For steady-state error, the numerator of the transfer function becomes important: for

$$\hat{G}(s) = \frac{n(s)}{d(s)}$$

Steady state error of $\hat{G}$ is

$$\begin{aligned} e_{ss} = \lim_{s\to 0}(1-\hat{G}(s))s\hat{u}(s) &= \lim_{s\to 0}\left(\frac{d(s)}{d(s)} - \frac{n(s)}{d(s)}\right)s\hat{u}(s) \\ &= \lim_{s\to 0}\frac{d(s)-n(s)}{d(s)}s\hat{u}(s) \end{aligned}$$

$\hat{u}(s)$ is the test signal

- **Step Input:** $s\hat{u}(s) = 1$
- **Ramp Input:** $s\hat{u}(s) = \frac{1}{s}$
- **Parabolic Input:** $s\hat{u}(s) = \frac{1}{s^2}$

Error Signals for Systems in Feedback

Use $\hat{G}(s) = \frac{n(s)}{d(s)}$

Lower Feedback Interconnection: $\dfrac{\hat{G}(s)\hat{K}(s)}{1+\hat{G}(s)\hat{K}(s)} = \dfrac{n(s)\hat{K}(s)}{d(s)+n(s)\hat{K}(s)}$

SS error for Lower Feedback Interconnection:

$$\hat{e}(s) = \left(1 - \frac{\hat{G}(s)\hat{K}(s)}{1+\hat{G}(s)\hat{K}(s)}\right) s\hat{u}(s) = \left(\frac{1}{1+\hat{G}(s)\hat{K}(s)}\right) s\hat{u}(s)$$

Step Response:

$$e_{ss,step} = \lim_{s\to 0} \frac{1}{1+\hat{G}(s)\hat{K}(s)} = \lim_{s\to 0} \frac{d(s)}{d(s)+n(s)\hat{K}(s)}$$

Ramp Response:

$$e_{ss,ramp} = \lim_{s\to 0} \frac{1}{1+\hat{G}(s)\hat{K}(s)} \frac{1}{s} = \lim_{s\to 0} \frac{d(s)}{d(s)+n(s)\hat{K}(s)} \frac{1}{s}$$

Parabolic Response:

$$e_{ss,parabola} = \lim_{s\to 0} \frac{1}{1+\hat{G}(s)\hat{K}(s)} \frac{1}{s^2} = \lim_{s\to 0} \frac{d(s)}{d(s)+n(s)\hat{K}(s)} \frac{1}{s^2}$$

Proportional Control Can Make Ramp Response Worse!!!

Consider the Suspension Example: **Open Loop:**

$$\hat{G}(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + s + 1}$$

$$1 - \hat{G}(s) = \frac{s^4 + 2s^3 + 3s^2 + s + 1 - s^2 - s - 1}{s^4 + 2s^3 + 3s^2 + s + 1} = \frac{s^4 + 2s^3 + 2s^2}{s^4 + 2s^3 + 3s^2 + s + 1}$$

Ramp Response:

$$\lim_{s \to 0} \frac{1 - \hat{G}(s)}{s} = \lim_{s \to 0} \frac{s^3 + 2s^2 + 2s}{s^4 + 2s^3 + 3s^2 + s + 1} = 0$$

What happens when we close the loop?

Closed Loop Transfer Function:

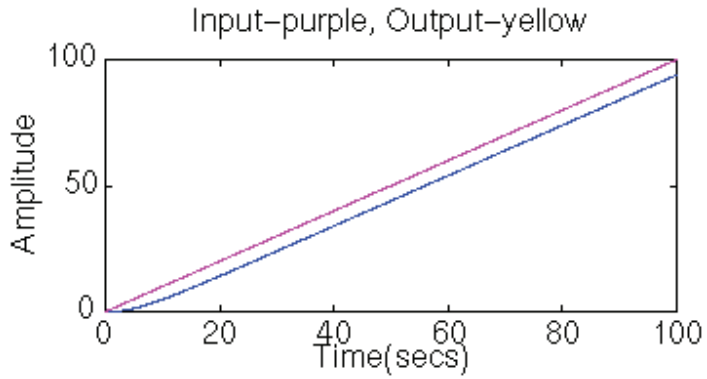
$$\frac{k(s^2 + s + 1)}{s^4 + 2s^3 + (3 + k)s^2 + (1 + k)s + (1 + k)}$$

Ramp Response:

$$e_{ss,ramp} = \lim_{s \to 0} \frac{1}{s(1 + \hat{G}(s)\hat{K}(s))} \cong \lim_{s \to 0} \frac{s^4 + 2s^3 + 3s^2 + s + 1}{k(s^2 + s + 1)} \frac{1}{s} = \infty$$

Proportional response isn't capable of controlling a ramp input

Example of Ramp Response



The only way to control a ramp input using feedback is to put a pole at the origin:

Controller: $\hat{K}(s) = \frac{1}{T_I s}$

Ramp Response:

$$e_{ss,ramp} = \lim_{s\to 0} \frac{d(s)}{d(s) + n(s)\hat{K}(s)} \frac{1}{s} = \lim_{s\to 0} \frac{d(s)}{sd(s)T_I + n(s)} \frac{T_I s}{s} = \frac{d(0)}{n(0)} T_I$$

By including $1/s$ in the controller, the steady-state error becomes finite.